

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2019 – 2020)

NEET : PRACTICE PAPER**TARGET : PRE-MEDICAL - 2020****Test Type : PT # 03****Test Pattern : NEET****TEST DATE : 03-04-2020*****Important Instructions*****Do not open this Test Booklet until you are asked to do so**

1. A seat marked with Reg. No. will be allotted to each student. The student should ensure that he/she occupies the correct seat only. If any student is found to have occupied the seat of another student, both the students shall be removed from the examination and shall have to accept any other penalty imposed upon them.
2. Duration of Test is **3 Hours** and Questions Paper Contains **180** Questions. The Max. Marks are **720**.
3. Student can not use log tables and calculators or any other material in the examination hall.
4. Student must abide by the instructions issued during the examination, by the invigilators or the centre incharge.
5. Before attempting the question paper ensure that it contains all the pages and that no question is missing.
6. Each correct answer carries 4 marks, while **1 mark will be deducted for every wrong answer**. Guessing of answer is harmful.
7. A candidate has to write his / her answers in the OMR sheet by darkening the appropriate bubble with the help of **Blue / Black Ball Point Pen only** as the correct answer(s) of the question attempted.
8. **Use of Pencil is strictly prohibited.**

Note : In case of any Correction in the test paper, please mail to dlpcorrections@allen.ac.in within 2 days along with **Paper code** and Your **Form No.**

Your Target is to secure Good Rank in Pre-MedicalCorporate Office : **ALLEN** CAREER INSTITUTE, "SANKALP", CP-6, Indra Vihar, Kota (Rajasthan)-324005☎ +91-744-5156100 ✉ info@allen.ac.in 🌐 www.allen.ac.in

TOPIC COVER

PHYSICS : Electrostatics + Capacitance

CHEMISTRY : Phenol, Ether + Aldehyde, Ketones + Carboxylic acid & derivatives + Amines & Anilene

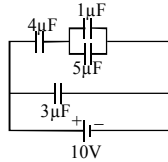
BIOLOGY : Plant Physiology (Complete) + Animal Tissue + Digestive system + Respiratory system + Excretory system + Circulatory system + Taxonomy , Biological Classification

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

1. For the circuit shown in the adjoining figure, the charge on $4\mu\text{F}$ capacitor is

- (1) $30\mu\text{C}$
(2) $40\mu\text{C}$
(3) $24\mu\text{C}$
(4) $54\mu\text{C}$



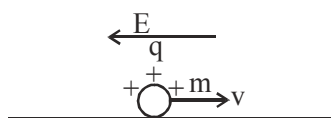
2. For two charge particles separated at a distance 3 cm, minimum electric force between them is nearly:

- (1) $e^2 \times 10^{+10} \text{ N}$ (2) $e^2 \times 10^{+11} \text{ N}$
(3) $e^2 \times 10^{+12} \text{ N}$ (4) $e^2 \times 10^{+13} \text{ N}$

3. In an electric field, an electron at rest, when placed, moves

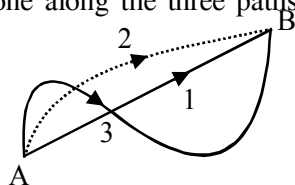
- (1) Along electric field
(2) Opposite to electric field
(3) Perpendicular to electric field
(4) Does not move

4. A particle of mass 'm' and charge q is placed in uniform electric field as shown figure. It is given the initial velocity v is opposite to the direction of electrical field. After how much time it will return to its original position



- (1) $\frac{mv}{Eq}$ (2) $\frac{mv}{2Eq}$ (3) $\frac{2mv}{Eq}$ (4) $\frac{mv}{3Eq}$

5. In the region of an electric field a charge is moved from point A to B via three different paths as shown in figure. W_1 , W_2 and W_3 denote the work done along the three paths. Then,

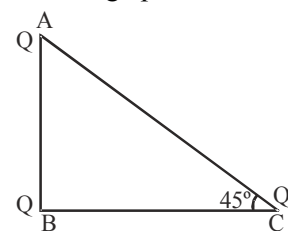


- (1) $W_1 < W_2 < W_3$ (2) $W_1 = W_2 < W_3$
(3) $W_1 < W_2 = W_3$ (4) $W_1 = W_2 = W_3$

6. There is an electric field E in x-direction. If the work done in moving a charge 0.2 C through a distance of 2 metres along a line making an angle 60° with the x-axis is 4 J, what is the value of E ?

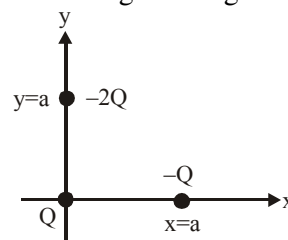
- (1) 4 N/C (2) 8 N/C
(3) $\sqrt{3}$ N/C (4) 20 N/C

7. If the force, on the charge placed at the corner A of an isosceles triangle, due to C, is F then the net force on the charge placed at B will be



- (1) $2\sqrt{2} F$ (2) $2F$ (3) $\frac{F}{2}$ (4) F

8. In given charge configuration angle made by net force on charge at origin from x-axis is :-



- (1) $\frac{\pi}{4}$ (2) $\tan^{-1}(2)$

- (3) $\tan^{-1}(1/2)$ (4) $\tan^{-1}(\sqrt{2})$

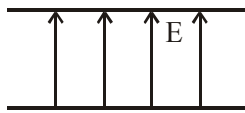
9. n number of identical spherical droplets, each having surface charge density σ coalesce to form a bigger drop of surface charge density

- (1) $(n)^{1/3} \sigma$ (2) $n \sigma$
(3) $(n)^{2/3} \sigma$ (4) $(n)^{5/3} \sigma$

10. In a regular polygon of n sides, each corner is at a distance r from the centre. Identical charges of magnitude q are placed at (n - 1) corners. The field at the centre is

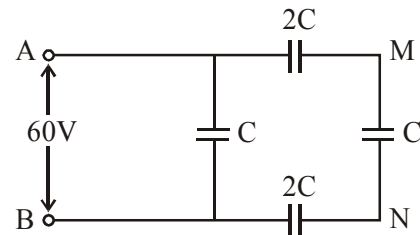
- (1) Zero (2) $\frac{kq}{r^2}$
(3) $\frac{(n-1)kq}{r^2}$ (4) $\frac{n-1}{n} \frac{kq}{r^2}$

11. An electric charge q exerts a force F on a similar electric charge q separated by a distance ' r '. A third charge $\frac{q}{4}$ is placed midway between the two charge. Now the force between previous two charge
- (1) $\frac{F}{3}$ (2) $\frac{F}{9}$ (3) Zero (4) F
12. An electron falls through a distance of 1.5 cm in a uniform electric field of magnitude 2.0×10^4 N/C as shown in the figure. The time taken by electron to fall through this distance is ($m_e = 9.1 \times 10^{-31}$ kg) (Neglect gravity)



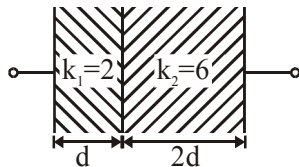
- (1) 1×10^{-9} s (2) 2.9×10^{-9} s
(3) 2.9×10^{-8} s (4) 1×10^{-8} s
13. The capacitance of a spherical capacitor with inner sphere of radius a and outer sphere of radius b is (Outer sphere is earthed)
- (1) $\frac{4\pi\epsilon_0 ab}{b-a}$ (2) $\frac{4\pi\epsilon_0 a^2}{b-a}$
(3) $\frac{4\pi\epsilon_0 b^2}{b-a}$ (4) $\frac{4\pi\epsilon_0 ab}{b+a}$
14. Two conducting spheres having radii r_1 and r_2 and surface charge densities σ_1 and σ_2 when connected to each other, if the potential energy of the system doesn't change then
- (1) $\frac{\sigma_1}{\sigma_2} = \frac{r_1}{r_2}$ (2) $\frac{\sigma_1}{\sigma_2} = \left(\frac{r_1}{r_2}\right)^2$
(3) $\frac{\sigma_1}{\sigma_2} = \frac{r_2}{r_1}$ (4) $\frac{\sigma_1}{\sigma_2} = \left(\frac{r_2}{r_1}\right)^2$
15. Two capacitors having capacitance C & $2C$ are charged to potential $4V$ and V respectively and then they are connected to each other. The loss in potential energy of the system is
- (1) CV^2 (2) $2CV^2$
(3) $3CV^2$ (4) $4CV^2$

16. In a parallel-plate capacitor of capacitance ' C ', a metal sheet is inserted between the plates, parallel to them. The thickness of the sheet is half of the separation between the plates. The capacitance now becomes
- (1) $4C$ (2) $2C$ (3) $C/2$ (4) $C/4$
17. In the circuit shown a potential difference of 60V is applied across AB. The potential difference between the points M and N is

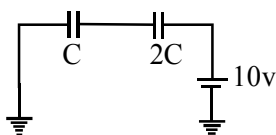


- (1) 10V (2) 15V (3) 20V (4) 30V
18. A parallel plate capacitor with air between the plates has a capacitance of 8 pF. If the distance between the plates is reduced by half, and the space between them is filled with a substance of dielectric constant 6, the capacitance now becomes
- (1) 90 pF (2) 96 pF
(3) 24 pF (4) Remains same
19. Two condensers of capacities $2C$ and C are joined in parallel and charged upto potential V . Then the battery is removed and the condenser of capacity C is filled completely with a medium of dielectric constant K . The potential difference across the capacitors will now be
- (1) $\frac{3V}{K+2}$ (2) $\frac{3V}{K}$ (3) $\frac{V}{K+2}$ (4) $\frac{V}{K}$
20. Two condensers of capacity $0.3\mu\text{F}$ and $0.6\mu\text{F}$ respectively are joined in series. The combination is connected across a potential difference of 6 volts. The ratio of energies stored by the condensers will be
- (1) 1/2 (2) 2 (3) 1/4 (4) 4

21. A capacitor of $1\mu\text{F}$ can withstand a maximum voltage of 6 kilovolt while another capacitor of $2\mu\text{F}$ can withstand a maximum voltage of 4 kilovolt. If the two capacitors are connected in series, the system will withstand a maximum voltage of
(1) 2kV (2) 4 kV (3) 6kV (4) 9 kV
22. A parallel plate capacitor is given a charge Q . E is the magnitude of electric field between the plates. The force on each plate of parallel plate capacitor is
(1) $\frac{1}{2}QE$ (2) QE (3) $\frac{1}{4}QE$ (4) $\frac{Q}{2E}$
23. A capacitor of capacitance C is charged to a potential V . The flux of the electric field through a closed surface enclosing the capacitor is
(1) $\frac{CV}{\epsilon_0}$ (2) $\frac{2CV}{\epsilon_0}$ (3) $\frac{CV}{2\epsilon_0}$ (4) Zero
24. A parallel plate capacitor has two layers of dielectrics as shown in figure. This capacitor is connected across a battery, then the ratio of potential difference across the dielectric layers is



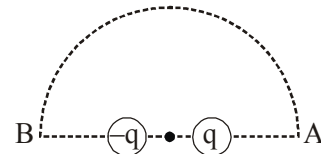
- (1) 4/3 (2) 1/2 (3) 1/3 (4) 3/2
25. In the circuit shown in figure $C = 6\mu\text{F}$. The charge stored in capacitor of capacity C is



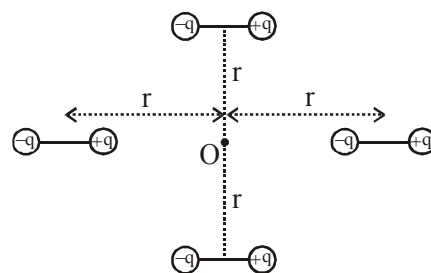
- (1) Zero (2) $90\mu\text{C}$ (3) $40\mu\text{C}$ (4) $60\mu\text{C}$
26. An electric dipole is placed in an uniform electric field with the dipole axis making an angle θ with the direction of the electric field. The orientation (θ) of the dipole for stable equilibrium is

- (1) $\frac{\pi}{6}$ (2) $\frac{\pi}{3}$ (3) 0 (4) $\frac{\pi}{2}$

27. Energy of a conducting sphere of radius R and charge q is $\frac{q^2}{8\pi\epsilon_0 R}$. This energy is stored in
(1) Both outside and inside the sphere.
(2) Only at the surface of the sphere.
(3) Uniformly outside the sphere starting from surface.
(4) Non-uniformly outside the sphere starting from surface.
28. Figure shows a semicircular path of very large radius concentric with a very short dipole. Magnitude of electric field (E) and electric potential (V) over that path in moving from A to B becomes :

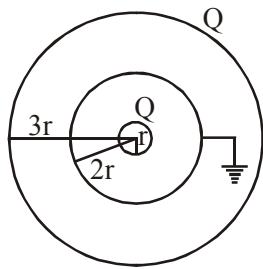


- (1) E initially decreases and then increases while V continuously decreases.
(2) E initially decreases and then increases while V continuously increases.
(3) E continuously decrease while V first increases then decreases.
(4) Both E and V initially decreases then increases.
29. Net electric field at the given point O due to all the four identical short electric dipoles of dipole moment p each shown in the figure is $\left(k = \frac{1}{4\pi\epsilon_0} \right)$



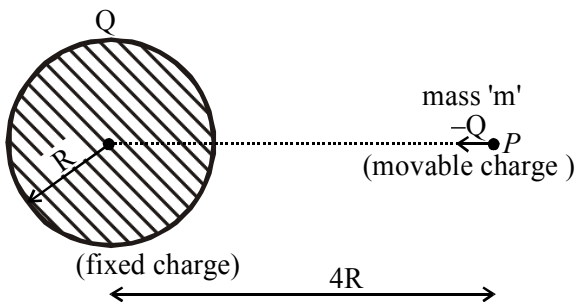
- (1) $\frac{kp}{r^3}$ (2) $\frac{2kp}{r^3}$ (3) $\frac{kp}{2r^3}$ (4) $\frac{4kp}{r^3}$

30. If the charge on the outermost and innermost shell is 'Q', then charge on the outer surface of middle shell is



- (1) $-\frac{5}{3}Q$ (2) $\frac{5}{3}Q$ (3) $-\frac{2}{3}Q$ (4) $\frac{2}{3}Q$

31. Velocity of the charge '-Q' when it strikes the surface of the fixed solid sphere is (Assume that charge -Q is released from rest from point P)

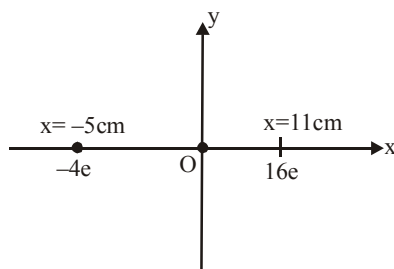


- (1) $\sqrt{\frac{3kQ^2}{2mR}}$ (2) $\sqrt{\frac{3kQ^2}{mR}}$
(3) $\sqrt{\frac{kQ^2}{mR}}$ (4) $\sqrt{\frac{2kQ^2}{3mR}}$

32. What happens to electrostatic potential energy of a two electron system, if one electron is brought towards another electron?

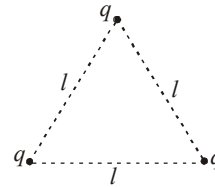
- (1) It becomes zero (2) It decreases
(3) It increases (4) It remains same

33. In given situation find position of equilibrium point :-



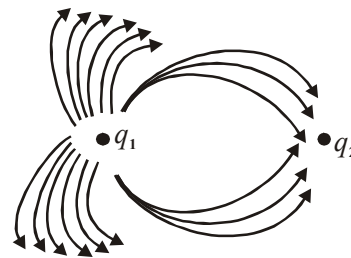
- (1) $x = -16$ cm (2) $x = -21$ cm
(3) $x = 11$ cm (4) $x = -18$ cm

34. Three charges q each are placed at the vertices of an equilateral triangle of side l . Find the work done to displace them to the vertices of an equilateral triangle of side $l/2$ is



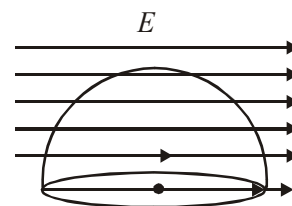
- (1) $\frac{q^2}{4\pi\epsilon_0 l}$ (2) $\frac{3q^2}{4\pi\epsilon_0 l}$
(3) $\frac{3q^2}{8\pi\epsilon_0 l}$ (4) None of these

35. Determine the ratio $\left|\frac{q_1}{q_2}\right|$, as lines of forces of two point charges are shown in figure



- (1) $\left|\frac{q_1}{q_2}\right| = 4$ (2) $\left|\frac{q_1}{q_2}\right| = 3$
(3) $\left|\frac{q_1}{q_2}\right| = 2$ (4) $\left|\frac{q_1}{q_2}\right| = 1$

36. In a region of uniform electric field E , a hemispherical surface is considered in such a way that field is parallel to its base (as shown in figure). The flux linked with the curved surface is

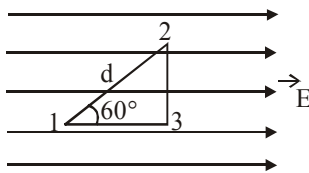


- (1) Zero (2) $-\pi R^2 E$
(3) $\pi R^2 E$ (4) $\frac{\pi R^2}{2} E$

37. A metal sphere A of radius a is charged to potential V . What will be its potential, if it is enclosed by a spherical conducting shell B of radius b and the two are connected by a wire?

(1) $\frac{a}{b}V$ (2) $\frac{b}{a}V$ (3) $\frac{b^2}{a}V$ (4) $\frac{a^2}{b}V$

38. The electric field $\vec{E}$ is constant in both magnitude and direction. Consider a path of length d at angle $\theta = 60^\circ$ with respect to field lines shown in figure. The potential difference between points 1 and 2 is



(1) $\frac{E}{d \cos 60^\circ}$ (2) $E d \cos 60^\circ$
(3) $\frac{Ed}{\cos 60^\circ}$ (4) $\frac{E}{d} \cos 60^\circ$

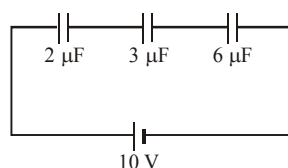
39. A dipole of electric dipole moment $\vec{P}$ is placed in a uniform electric field of strength $\vec{E}$. If θ is the angle between positive directions of $\vec{P}$ and $\vec{E}$, the potential energy of the electric dipole is maximum when θ is

(1) Zero (2) $\frac{\pi}{2}$ (3) π (4) $\frac{\pi}{4}$

40. Out of two concentric conducting spherical shells, the inner one is given some charge and the outer one is earthed. The energy of the system is stored in

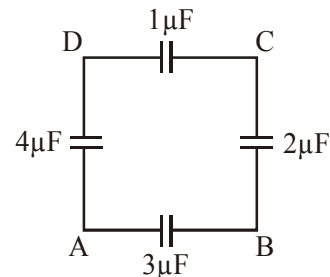
- (1) inside the inner sphere
(2) outside the outer sphere
(3) the space between both spheres
(4) Both (1) & (2)

41. In the figure, the charge on $3 \mu\text{F}$ capacitor is



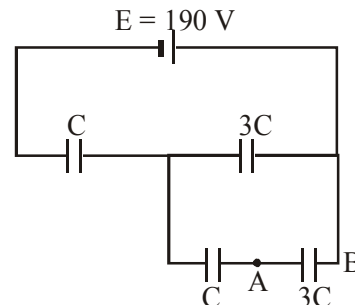
(1) $5 \mu\text{C}$ (2) $10 \mu\text{C}$ (3) $3 \mu\text{C}$ (4) $6 \mu\text{C}$

42. Four capacitors are connected to each other as shown in figure. The ratio of equivalent capacitance between A & B and that of between D & C is



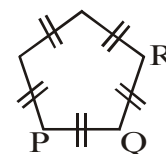
(1) $\frac{25}{7}$ (2) $\frac{13}{7}$ (3) $\frac{25}{13}$ (4) $\frac{1}{3}$

43. In the circuit shown in figure, potential difference between A and B is



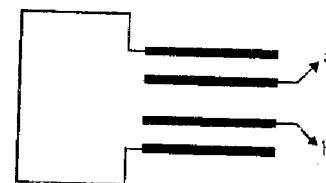
(1) 60 V (2) 10 V
(3) 30 V (4) 90 V

44. Five capacitors, each of capacitance value C are connected as shown in the figure. The ratio of capacitance between P and R, and the capacitance between P and Q, is

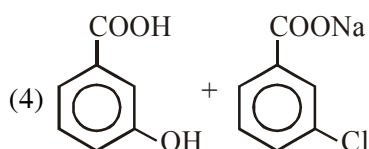
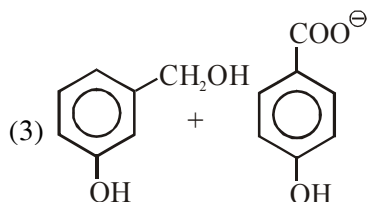
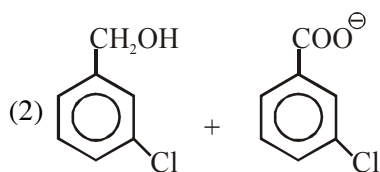
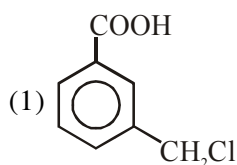
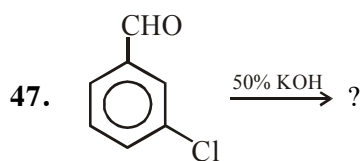
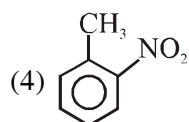
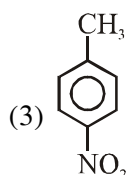
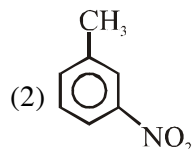
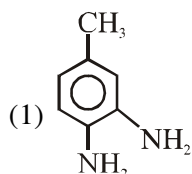
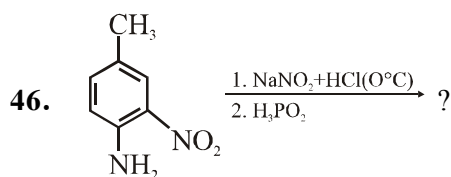


(1) 3 : 1 (2) 5 : 2 (3) 2 : 3 (4) 1 : 1

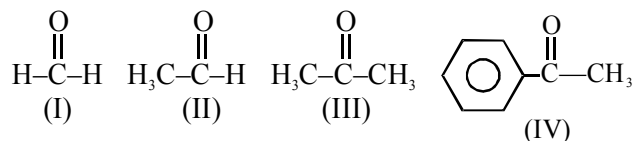
45. Four metallic plate each with a surface area of A are placed at a distance d from each other. The plates are connected as shown in the circuit diagram. Then the capacitance of the system between a and b is



(1) $\frac{3\epsilon_0 A}{d}$ (2) $\frac{2\epsilon_0 A}{d}$
(3) $\frac{2\epsilon_0 A}{3d}$ (4) $\frac{3\epsilon_0 A}{2d}$

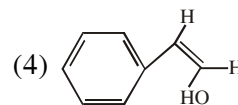
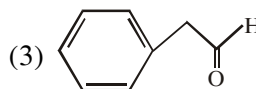
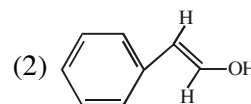
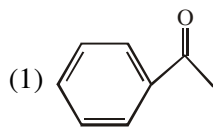
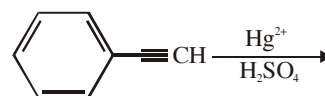


48. Arrange the following compounds in decreasing order of nucleophilic addition reaction :-



- (1) II > IV > III > I (2) I > II > III > IV
 (3) IV > III > II > I (4) II > III > IV > I

49. What is the major organic product of the following reaction ?

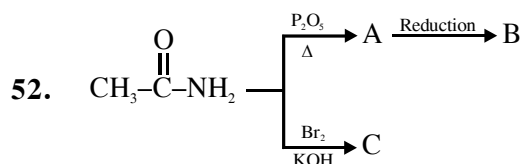


50. The reagent that can distinguish benzaldehyde from propionaldehyde is :-

- (1) I_2/NaOH
 (2) Fehling's solution
 (3) Tollens reagent
 (4) 2, 4-dinitrophenyl hydrazine

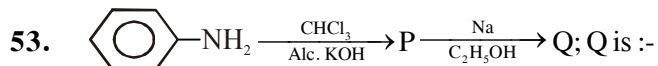
51. 2-Pentanone and 3-Pentanone can be distinguished by ?

- (1) DNP (2) Tollen reagent
 (3) I_2/NaOH (4) All above

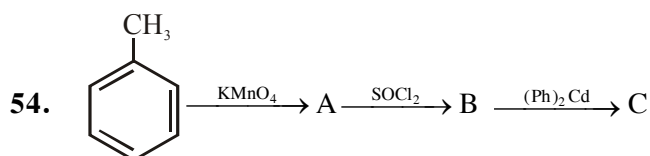


B and C are :-

- (1) Structural isomers (2) Homologues
 (3) Identical (4) Stereoisomers



- (1) c1ccc(O)cc1 (2) c1ccc(CN)cc1
(3) c1ccc(NC)cc1 (4) c1ccc(N)cc1CC



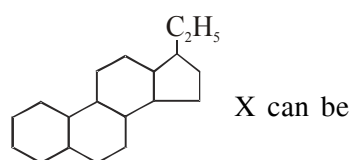
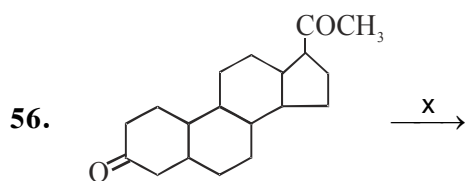
Which of the following statement will be correct for 'C' ?

- (1) 'C' give positive haloform test
(2) 'C' give white precipitate with NaHSO_3
(3) 'C' on reaction with LiAlH_4 form 2° alcohol
(4) 'C' reacts with HCN form racemic mixture product



The compound x is :-

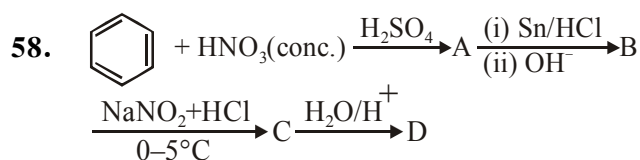
- (1) CC(=O)N (2) CCC(=O)N
(3) CC (4) CCN



- (1) $\text{NH}_2 - \text{NH}_2 / \text{KOH}$
(2) $\text{Zn} - \text{Hg} / \text{HCl}$
(3) Red P + HI
(4) All

57. Primary and secondary amines can be distinguished by :-

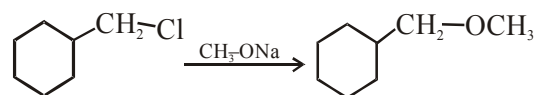
- (1) Hoffmann's mustard oil reaction
(2) carbylamine reaction
(3) Hinsberg reagent
(4) All



What is D-

- (1) Nc1ccc(C(=O)O)cc1 (2) Oc1ccc(N)cc1
(3) c1ccccc1 (4) Clc1ccccc1

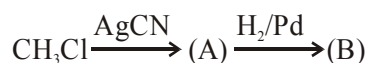
59. Consider the following reaction -



The mechanism of reaction will be :-

- (1) SN^1 (2) SN^2 (3) E_2 (4) E_1

60. In the chemical reaction



What is the structure of B :-

- (1) CCN (2) CCNC
(3) CC(=O)N (4) CCC

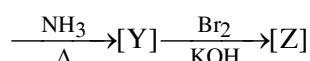
61. Which of the following give Bad smell when treated with $\text{CHCl}_3 / \text{KOH}$?

- (1) CC(C)N (2) CCNC
(3) c1ccccc1NC (4) C1CCNCC1

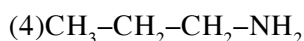
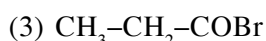
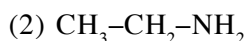
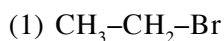
62. Which of the following will not give D.N.P. test?



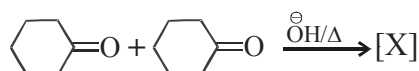
63. Starting from propanoic acid the following reactions were carried out as following



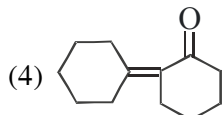
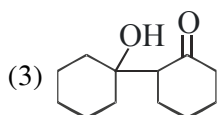
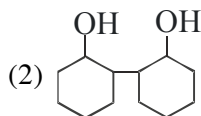
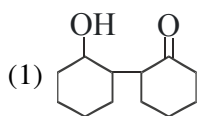
Product [Z] will be?



64. In the reaction :



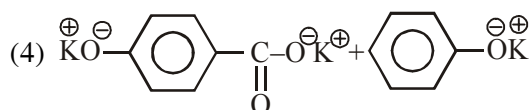
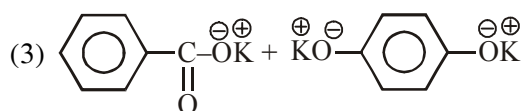
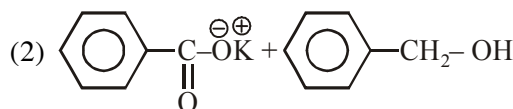
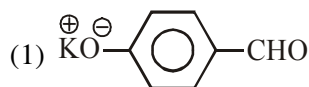
Product [X] will be :



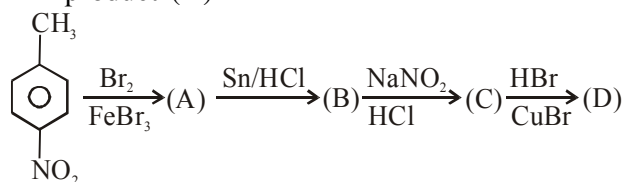
65. Which of the following is most reactive for nucleophilic substitution reaction ?



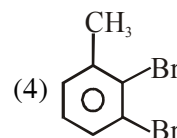
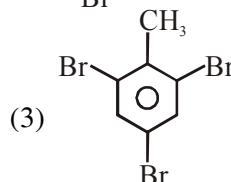
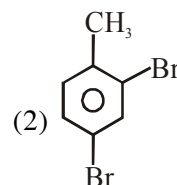
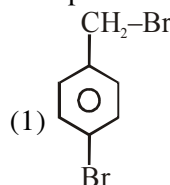
66. Which product is formed when Ph-CHO is treated with concentrated KOH solution :-



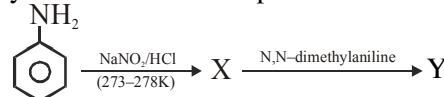
67. In a set of reactions p-nitrotoluene yielded a product (D)



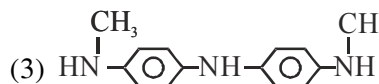
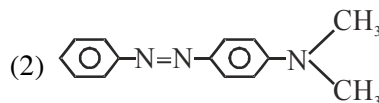
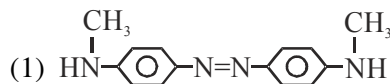
The product (D) would be :-



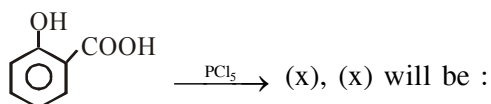
68. Aniline in a set of the following reactions yielded a coloured product 'Y'



The structure of 'Y' would be :-



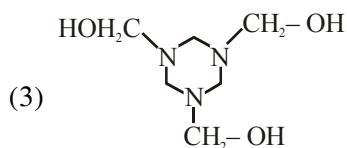
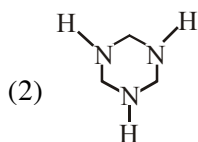
69. In the given reaction



- (1)
 (2)
 (3)
 (4)

70. Formaldehyde reacts with ammonia to give :

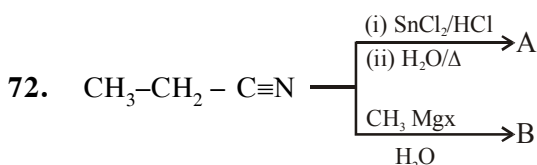
- (1) $\text{H}_2\text{C} = \text{NH}$



- (4) Hexamethylene tetramine

71. $\text{R}-\text{COOH} \xrightarrow[\text{(ii) H}_3\text{O}^+]{\text{(i) LiAlH}_4/\text{ether}}$ Product will be :-

- (1) $\text{R}-\text{CHO}$ (2) $\text{R}-\text{CH}_2-\text{OH}$
 (3) $\text{R}-\text{CH}_3$ (4) $\text{R}-\text{C}(=\text{O})-\text{R}$



A and B are :-

- (1) $\text{CH}_3\text{CH}_2\text{CHO}$, $\text{CH}_3\text{CH}_2-\text{C}(=\text{O})-\text{CH}_3$
 (2) CH_3CHO , CH_3COCH_3
 (3) $\text{CH}_3\text{CH}_2\text{CHO}$, CH_3CHO
 (4) CH_3CHO , $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$

73. Which does not give aldol condensation :-

- (1)
 (2) CH_3CHO
 (3) $\text{CH}_3\text{CH}_2\text{CHO}$
 (4) $\text{C}_6\text{H}_5\text{CH}_2\text{CHO}$

74. Which of the following acid has lowest value of dissociation constant :-

- (1) $\text{CH}_3-\underset{\text{F}}{\text{CH}}-\text{COOH}$
 (2) $\underset{\text{F}}{\text{CH}_2}-\text{CH}_2-\text{COOH}$
 (3) $\underset{\text{Br}}{\text{CH}_2}-\text{CH}_2-\text{COOH}$
 (4) $\text{CH}_3-\underset{\text{Br}}{\text{CH}}-\text{COOH}$

75. $\text{Ph}-\text{C}\equiv\text{CH} \xrightarrow[\text{Hg}^{2+}/\text{H}^+]{\text{H}_2\text{O}}$ Product is :-

- (1) PhCHO
 (2) $\text{Ph}-\text{C}(=\text{O})-\text{OH}$
 (3) $\text{Ph}-\text{C}(=\text{O})-\text{CH}_3$
 (4) None

76. $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\text{CH}_2-\text{NH}_2 \xrightarrow[\text{HCl}]{\text{NaNO}_2}$ Main Product is :-

- (1) $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\text{CH}_2-\text{OH}$
 (2) $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\text{CH}_3$
 (3) $\text{CH}_3-\underset{\text{CH}_3}{\text{C}}(\text{OH})-\text{CH}_3$
 (4) None

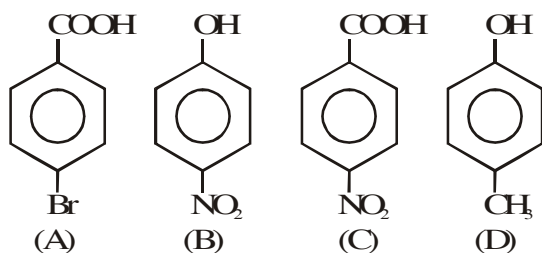
77. $\text{CH}_3-\underset{\text{OH}}{\text{CH}}-\text{CH}_3 \xrightarrow{\text{I}_2+\text{Ca}(\text{OH})_2}$ Product(s) :-

- (1) $\text{CHI}_3+\text{HCOOCa}$
 (2) $\text{CHI}_3+(\text{HCOO})_2\text{Ca}$
 (3) $\text{CHI}_3+\text{CH}_3\text{COOCa}$
 (4) $\text{CHI}_3+(\text{CH}_3\text{COO})_2\text{Ca}$

78. A mixture of benzaldehyde and formaldehyde on heating with aqueous NaOH solution gives :-

- (1) PhCH_2OH and HCOONa
- (2) PhCOONa and CH_3OH
- (3) PhCOONa and HCOONa
- (4) PhCH_2OH and CH_3OH

79. Arrange the following compounds in increasing order of their acidic strength.

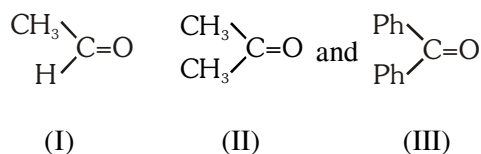


- (1) $(C) > (A) > (B) > (D)$
- (2) $(C) > (B) > (D) > (A)$
- (3) $(C) > (B) > (A) > (D)$
- (4) $(C) > (A) > (D) > (B)$

80. Salicylaldehyde is obtained when phenol is heated with CHCl_3 and aqueous NaOH. This reaction is known by which name ?

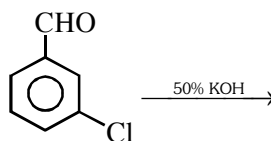
- (1) Reimer Tiemann Carboxylation
- (2) Hoffmann's reaction
- (3) Reimer Tiemann Formylation reaction
- (4) Kolbe Schmitt reaction

81. The order of reactivity of phenyl magnesium bromide (PhMgBr) with the following compounds :-



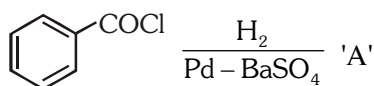
- (1) $I > II > III$
- (2) $III > II > I$
- (3) $II > I > III$
- (4) $I > III > II$

82. Predict the products in the given reaction.



- (1) +
- (2) +
- (3) +
- (4) +

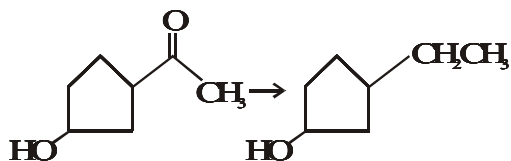
83. Consider the following reaction :



The product 'A' is :

- (1) $\text{C}_6\text{H}_5\text{COCH}_3$
- (2) $\text{C}_6\text{H}_5\text{Cl}$
- (3) $\text{C}_6\text{H}_5\text{CHO}$
- (4) $\text{C}_6\text{H}_5\text{OH}$

84. The appropriate reagent for the transformation

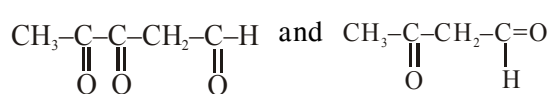


- | | |
|----------------------------|---|
| (1) Zn(Hg), HCl | (2) $\text{NH}_2\text{NH}_2, \text{OH}^-$ |
| (3) H_2/Ni | (4) NaBH_4 |

85. When calcium formate and calcium acetate is heated then major product will be :-

- (1) CH_3COCH_3
- (2) $\text{CH}_3-\text{C}(=\text{O})-\text{OCH}_3$
- (3) HCHO
- (4) $\text{CH}_3-\text{C}(=\text{O})-\text{H}$

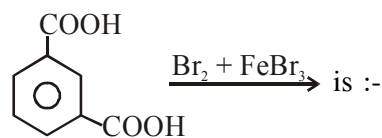
86. A compound having molecular formula $\text{C}_{10}\text{H}_{14}$ is hydrogenated with H_2/Pd to give 1-isopropyl-4-methyl cyclohexane and on reductive ozonolysis it give $\text{H}-\text{C}(=\text{O})-\text{H}$,



The structure of the compound would be :

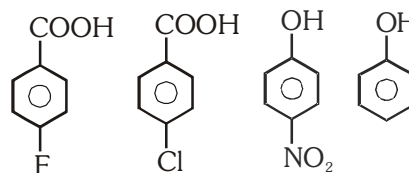
- (1)
- (2)
- (3)
- (4)

87. The major product in the reaction



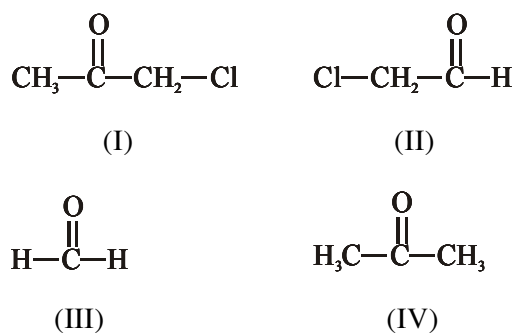
- (1)
- (2)
- (3)
- (4)

88. Order of acidic strength for the following :



- (1) $\text{II} > \text{I} > \text{III} > \text{IV}$
- (2) $\text{I} > \text{II} > \text{III} > \text{IV}$
- (3) $\text{II} > \text{I} > \text{IV} > \text{III}$
- (4) $\text{IV} > \text{I} > \text{III} > \text{II}$

89. Arrange the compounds in order of decreasing reactivity for nucleophilic addition reaction :

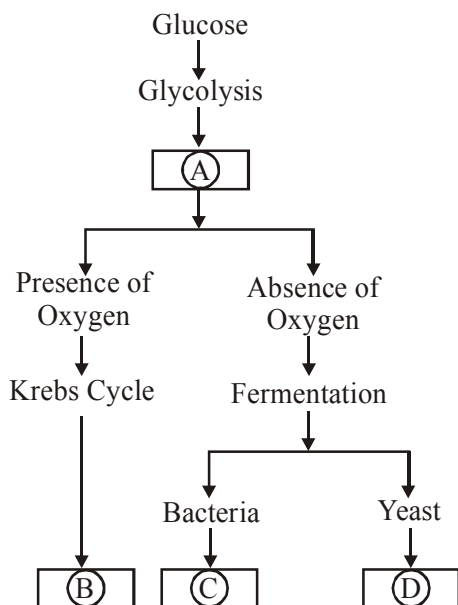


- (1) $\text{I} > \text{IV} > \text{II} > \text{III}$
- (2) $\text{I} > \text{II} > \text{III} > \text{IV}$
- (3) $\text{II} > \text{III} > \text{I} > \text{IV}$
- (4) $\text{II} > \text{I} > \text{III} > \text{IV}$

90. Which of following amides does not undergo hoffmann bromamide reaction :-

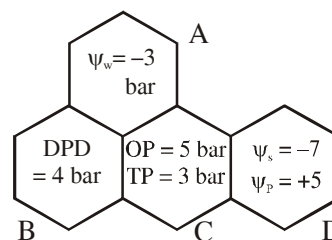
- (1) CH_3CONH_2
- (2) $\text{C}_6\text{H}_5\text{CONH}_2$
- (3) $\text{C}_6\text{H}_5\text{CH}_2\text{CONH}_2$
- (4) $\text{C}_6\text{H}_5\text{CONHCH}_3$

91. The following is a simplified scheme showing the fate of glucose during aerobic and anaerobic respiration. Identify the end products that are formed at stages indicated as A, B, C and D. Identify the correct option from those given below:-



- (1) A = Carbondioxide and water, B = pyruvic acid, C = ethyl alcohol and carbondioxide, D = lactic acid
 (2) A = pyruvic acid, B = carbondioxide, C = ethyl alcohol and carbondioxide, D = lactic acid
 (3) A = pyruvic acid, B = carbondioxide, C = lactic acid, D = ethyl alcohol and carbondioxide
 (4) A = pyruvic acid, B = ethyl alcohol and carbondioxide, C = lactic acid, D = carbondioxide and water
92. In aerobic respiration, one glucose produces:-
 (1) $8 \text{ NADH}_2 + 2 \text{ FADH}_2 + 24 \text{ ATP}$
 (2) $12 \text{ NADH}_2 + 2 \text{ FADH}_2 + 38 \text{ ATP}$
 (3) $12 \text{ NADH}_2 + 30 \text{ ATP} + 12 \text{ H}_2\text{O} + 2 \text{ GTP}$
 (4) $10 \text{ NADH}_2 + 2 \text{ FADH}_2 + 2 \text{ ATP} + 2 \text{ GTP}$
93. Simple as well as facilitated diffusion both –
 (1) Require ATP energy
 (2) Show downhill transport
 (3) Require special membrane proteins
 (4) Can not transfer the substances having a hydrophobic moiety

94. On the basis of following diagram identify the correct direction of water movement



- (1) $A \leftarrow B \leftarrow C \leftarrow A$
 (2) $A \rightarrow B \rightarrow C \rightleftharpoons D$
 (3) $A \rightarrow B \leftarrow C \rightleftharpoons D$
 (4) $A \leftarrow B \rightarrow C \leftarrow D$
95. Movement of molecules across a membrane independent of other molecules is known as –
 (1) Coport (2) Uniport
 (3) Antiport (4) Symport
96. Which of the following is not associated with Guttation –
 (1) Minerals (2) H_2O
 (3) Organic Acids (4) Proteins
97. Older dying leaves export some minerals to younger leaves. This statement is –
 (1) Correct for all minerals
 (2) Not correct for elements that are structural components
 (3) Correct for N & P only
 (4) Not correct for Ca, K, S
98. Which of these is released by bacteria during symbiotic N_2 fixation –
 (1) Auxin
 (2) Haem of Leghaemoglobin
 (3) ATP for reduction of N_2
 (4) Globin of Leghaemoglobin
99. The simplest expression of arithmetic growth is exemplified by
 (1) Root elongation at increasing rate
 (2) Root elongation at constant rate
 (3) Early zygotic growth
 (4) All the above

100. Exponential growth can be expressed as –

(1) $L_t = L_0 + rt$

(2) $W_1 = W_0 e^{rt}$

(3) $\frac{dN}{dt} = rN$

(4) $\frac{dN}{dt} = rN \left[\frac{K - N}{K} \right]$

101. Select out the odd one with reference to the events which are affected by more than one PGR–

- (1) Dormancy of seeds/Buds
- (2) Abscission
- (3) Senescence
- (4) Bolting

102. During photoperiodic response, required hypothetic hormonal substance migrate –

- (1) From leaf to Root
- (2) From Root to shoot apices
- (3) From leaf to shoot apices
- (4) From shoot apices to leaf

103. In pernicious anaemia, failure of erythrocytes to mature is mainly due to lack of :-

- (1) Vitamin-K
- (2) Thiamine
- (3) Pyridoxine
- (4) Cyanocobalamine

104. Milk protein casein $\xrightarrow{(y)}$ paracasein $\xrightarrow{(M)}$ calcium paracaseinate (curding of milk) :-

In the above reaction, latter (y) and (M) denote:-

- (1) Rennin and Ca^{+2} respectively
- (2) Ca^{+2} and rennin respectively
- (3) Rennin, HCl and Ca^{+2} respectively
- (4) Pepsin and Ca^{+2} respectively

105. Read the following statements :-

- (a) Human liver is the largest endocrine gland of body and having two lobes
 - (b) Hepatic lobules are structural and functional unit of liver and contained hepatocytes which are arranged in chord like manner
 - (c) Glisson's capsule is covering of each lobules and made up by connective tissue
 - (d) Bile juice is formed and secreted by hepatocytes and is stored into liver sinusoid
- Out of these which statements are corrects and incorrects ?

- (1) Statements a and b are correct while c & d are incorrect
- (2) Statements a and d are correct while b & c are incorrect
- (3) Statements b & c are correct while a & d are incorrect
- (4) Statements b and d are correct while a & c are incorrect

106. If for some reason the parietal cells of the stomach epithelium become partially non-functional what is likely to happen ?

- (1) The pancreatic enzymes and specially the trypsin and lipase will not work efficiently
- (2) The pH of stomach will fall abruptly
- (3) Steapsins will be more effective
- (4) Proteins will not be adequately hydrolysed by pepsin into proteoses and peptones

107. Match column-A (epithelial tissue) with column-B (their location) :-

- | Column-A
(Epithelial tissue) | Column-B
(their location) |
|----------------------------------|------------------------------|
| (i) Simple squamous epithelium | (A) Trachea |
| (ii) Simple cuboidal epithelium | (B) Bile-duct |
| (iii) Simple columnar epithelium | (C) Bowman's Capsule |
| (iv) Pseudostratified epithelium | (D) Sweat gland |
| | (E) Buccal cavity |

Correct matching is :-

- (1) i-C, ii-D, iii-A, iv-B
(2) i-C, ii-D, iii-B, iv-A
(3) i-D, ii-C, iii-B, iv-A
(4) i-D, ii-A, iii-B, iv-C

108. Which of the following is function of the basement membrane?

- (1) Excitability
(2) Contractility
(3) Active transport
(4) Molecular filtering

109. Which among the following is not characteristics of yellow fibres of connective tissue ?

- (1) Occur singly
(2) Contain elastin
(3) Fewer in number
(4) Provide toughness and strength

110. Given below a table in which name of digestive enzymes, their substrate and their products are given in column I, II and III respectively. Select the option in which correctly matched pairs are given ?

S. No.	Column-I Name of enzyme	Column-II Substrate	Column-III Product
(i)	Maltase	Maltose	Glucose + Fructose
(ii)	Lactase	Lactose	Glucose + galactose
(iii)	Nucleosidase	Nucleotide	Sugar + N- Bases
(iv)	Sucrase	Sucrose	Glucose + Fructose
(v)	Nucleotidase	Nucleotide	Nucleoside + phosphate group

Options :-

- (1) (i), (ii), (iv) & (v) (2) (i), (ii), (iii) & (v)
(3) (ii), (iv) & (v) (4) (i), (iii) & (iv)

111. Find out the correct match from the following table :-

	Column-I Name of salivary gland	Column-II Their duct	Column-III Their property
(i)	Parotid gland	Stenson's duct	Largest salivary gland
(ii)	Sublingual gland	Wirsung duct	Smallest salivary gland
(iii)	Submandibular gland	Wharton's duct	Maximum salivation

Options :-

- (1) Only (ii)
(2) (i) and (iii)
(3) Only (i)
(4) (i), (ii) and (iii) all

112. Which one of the following is the correct match of the site of the action on the given substrate, the enzyme acting upon it and the end product?

- (1) Duodenum :→ Triglycerides $\xrightarrow{\text{Trypsin}}$ Monoglycerides
(2) Small intestine :→ protein $\xrightarrow{\text{Pepsin}}$ Amino acid
(3) Stomach :→ Nucleic acid $\xrightarrow{\text{Nuclease}}$ nucleotide
(4) Ileum :→ Dipeptides $\xrightarrow{\text{Dipeptidase}}$ Amino acid

113. Read the statement I and II :-

Statement-I :- The human small intestine is the longest part in alimentary canal.

Statement-II :- Absorption of digested food requires very large surface area.

Select the correct option on the basis of above statement ?

- (1) Statement I is correct but II is wrong
(2) Statement I and II both are wrong
(3) Statement I and II both are correct
(4) Statement II is correct but I is wrong

- 114.** What would happen when blood is acidic?
 (1) Binding of oxygen with haemoglobin is increases
 (2) Red blood corpuscles are formed in higher number
 (3) Binding of oxygen with haemoglobin is decreases
 (4) There is no change in oxygen binding nor number of RBC.
- 115.** Residual air mostly occurs in
 (1) alveoli (2) bronchus
 (3) nostrils (4) trachea
- 116.** Trachea is lined with incomplete rings of
 (1) fibrous cartilage (2) calcified cartilage
 (3) elastic cartilage (4) hyaline cartilage
- 117.** Amount of oxygen carried by one gram of haemoglobin is
 (1) 20 ml (2) 1.34 ml
 (3) 13.4 ml (4) None of these
- 118.** Gases diffuse over the respiratory surface because of
 (1) P_{O_2} is more in alveoli than in blood
 (2) P_{O_2} is more in blood than in tissues
 (3) P_{CO_2} is more in alveoli than in blood
 (4) P_{CO_2} is more in blood than in tissues
- 119.** Dead space is present in
 (1) respiratory tract
 (2) nasal chambers only
 (3) alveolar space
 (4) pleural cavity
- 120.** Which is the correct sequence of the air passage in man?
 (1) Nasal cavity → pharynx → trachea → larynx → bronchi → bronchioles → alveoli
 (2) Nasal cavity → pharynx → larynx → trachea → bronchi → bronchioles → alveoli
 (3) Nasal cavity → larynx → pharynx → trachea → bronchi → bronchioles → alveoli
 (4) Nasal cavity → larynx → bronchi → pharynx → trachea → bronchioles → alveoli
- 121.** Food and air pathways are divided at
 (1) larynx (2) pharynx
 (3) stomach (4) oesophagus
- 122.** The structure which does not contribute to the breathing movements in mammals is
 (1) rib
 (2) larynx
 (3) diaphragm
 (4) intercostal muscles
- 123.** Lining of trachea is made up of
 (1) stratified ciliated epithelium
 (2) pseudostratified ciliated epithelium
 (3) simple squamous epithelium
 (4) stratified cuboidal epithelium
- 124.** Which of the following statements is correct?
 (1) Inspiration is an active process
 (2) Inspiration is a passive process
 (3) Expiration is an active process
 (4) Both expiration and inspiration are passive processes
- 125.** During expiration the diaphragm becomes
 (1) normal (2) oblique
 (3) flattened (4) dome-shaped
- 126.** Rate of breathing in an adult human is
 (1) 10-12/min (2) 12-18/min
 (3) 20-25/min (4) 30-35/min
- 127.** Which of the following statements, best summarises the relationship between respiratory rate and body size in related animals?
 (1) Larger the animal higher the respiratory rate
 (2) Smaller the animal lower the respiratory rate
 (3) Smaller the animal higher the respiratory rate
 (4) Size and respiratory rate are not related in any fashion

- 128.** About 1200 ml of air is always known to remain inside the human lungs. It is described as
- (1) functional residual capacity
 - (2) residual volume
 - (3) expiratory reserve volume
 - (4) inspiratory reserve volume
- 129.** Vital capacity of lungs is
- (1) IRV + ERV
 - (2) IRV + ERV + TV
 - (3) IRV + ERV + TV – RV
 - (4) IRV + ERV + TV + RV
- 130.** How the transport of O_2 and CO_2 by blood happens?
- (1) With the help of RBCs and WBCs
 - (2) With the help of WBCs and blood serum
 - (3) With the help of platelets and plasma
 - (4) With the help of RBCs and blood plasma
- 131.** Which form of iron is found in haemoglobin?
- (1) Fe^{+2}
 - (2) Fe^{+3}
 - (3) In the form of molecule
 - (4) In the form of FeO
- 132.** How many molecules of oxygen can associate with a molecule of haemoglobin in man?
- (1) One (2) Two (3) Three (4) Four
- 133.** In which condition, oxygen dissociation curve of haemoglobin shift to right of normal curve?
- (1) Decrease in pH
 - (2) Decrease in acidity
 - (3) Decrease in temperature
 - (4) Decrease in CO_2 concentration
- 134.** Dissociation of oxyhaemoglobin can be promoted by
- (1) low P_{CO_2}
 - (2) high P_{CO_2}
 - (3) high blood pH
 - (4) low body temperature
- 135.** Which one represents pulmonary circulation?
- (1) Left auricle (oxygenated blood → lungs (deoxygenated blood → Right auricle
 - (2) Left auricle (deoxygenated blood → lungs (oxygenated blood → Right auricle
 - (3) Left auricle (oxygenated blood → lungs (deoxygenated blood → Left auricle
 - (4) Right auricle (deoxygenated blood → lungs (oxygenated blood → Left auricle
- 136.** Blood vessel carrying least CO_2 is
- (1) Pulmonary vein (2) Pulmonary artery
 - (3) Vena Cava (4) Hepatic vein
- 137.** Cardiac muscles are found in:
- (1) epicardium (2) myocardium
 - (3) endocardium (4) all of these
- 138.** Tricuspid valve is present between the
- (1) two atria
 - (2) two ventricles
 - (3) left atrium and ventricle
 - (4) right atrium and ventricle
- 139.** Which term does not apply to human heart?
- (1) Pacemaker (2) Mitral valve
 - (3) Neurogenic (4) Four chambered
- 140.** The duration of cardiac cycle in a normal man is
- (1) 0.8 seconds (2) 80 seconds
 - (3) 8 seconds (4) 72 seconds
- 141.** The "lubb" sound (first heart sound) is caused by the
- (1) filling of the ventricles
 - (2) ventricular contraction
 - (3) closing of the AV valves
 - (4) closing of semilunar valves
- 142.** Select ammonotelic from the following
- (1) Many bony fishes
 - (2) aquatic amphibians
 - (3) aquatic insects
 - (4) All

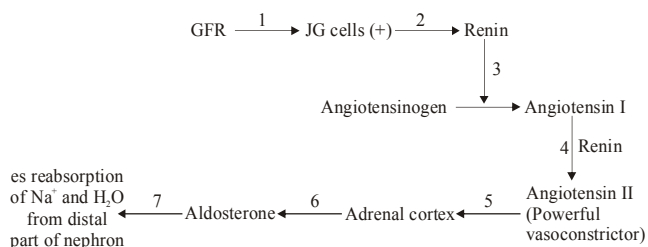
- 143.** Metabolic wastes containing nitrogen in our body are due to
 (1) Carbohydrates (2) Proteins
 (3) Fats (4) Vitamins
- 144.** Excretion involves process in which
 (1) harmful substances in the body are chemically changed
 (2) substances of no further use or those present in excessive quantities are thrown out of the body
 (3) harmful substances are stored in cells before being eliminated
 (4) expulsion of urine from the urinary bladder and sweat from the skin
- 145.** Ammonia which is produced by metabolism is converted intoA.... in the liver of mammals and released intoB... which is filtered andC.... out by kidney
 (1) A–Uric acid, B–Blood, C–excreted
 (2) A–Urea, B–Blood, C–excreted
 (3) A–Amino acid, B–Blood, C–Excreted
 (4) A–Sugar, B–Blood, C–excreted
- 146.** What gets increased in blood if liver becomes functionless ?
 (1) Urea (2) Ammonia
 (3) Uric acid (4) Proteins
- 147.** In aquatic organisms, the waste end product of nitrogen metabolism is
 (1) Urea (2) Nitrogen
 (3) Ammonia (4) Allantois
- 148.** Uric acid is excreted by
 (1) Pigeon (2) Frog
 (3) Rabbit (4) Man
- 149.** The least toxic nitrogenous waste is
 (1) Ammonia (2) Ammonia + Urea
 (3) Urea (4) Uric acid
- 150.** Excretory and osmoregulatory structure in cockroach is
 (1) flame cells
 (2) green glands
 (3) nephridia
 (4) malpighian tubules
- 151.** Kidneys in human situated between
 (1) T12–L3 (2) T11–L2
 (3) T12–L1 (4) T12–L5
- 152.** Part of Nephron is situated in cortex completely
 A. Malpighian Corpuscle
 B. PCT
 C. DCT
 D. Loop of Henle
 E. Collecting duct
 (1) A, B, C only (2) B, C only
 (3) A, B, C, D only (4) D, E only
- 153.** Mammalian kidney serve to excrete
 (1) excess water, urea and amino acids
 (2) excess salts, urea and excess water
 (3) excess salts, excess water and excess amino acids
 (4) excess salts, excess water and excess amino acids
- 154.** Part through which arteries and veins enter or leave the kidney is called
 (1) major calyces (2) minor calyces
 (3) hilus (4) renal pore
- 155.** The values of GFR in healthy individual is
 (1) 125 ml/min (2) 150 ml/min
 (3) 100 ml/min (4) 200 ml/min
- 156.** DCT helps in
 (1) conditional reabsorption of Na^+ and water
 (2) HCO_3^- absorption
 (3) pH maintenance
 (4) selective secretion of H^+ and K^+
- 157.** The effect of antidiuretic hormone (ADH) on the kidney is to increases the
 (1) excretion of water
 (2) excretion of Na^+
 (3) Permeability of the distal nephron to water
 (4) glomerular filtration rate

158. Arrange the following steps in order

- (1) excessive loss of fluid
- (2) stimulation of osmoreceptor
- (3) stimulation of Hypothalamus
- (4) release of ADH or Vasopressin
- (5) ADH facilitate water reabsorption from distal tubules
- (6) Increase in body fluid switch off osmoreceptor and suppress release of ADH

- (1) 1,2,3,4,5,6 (2) 1,2,3,4,6,5
- (3) 6,1,2,3,4,5 (4) 2,3,4,1,5,6

159. Select the incorrect from the following :



- (1) 2,3 (2) 1,3 (3) 4 (4) 5,6

160. Species is

- (1) Smallest unit of unit of classification
- (2) Specific unit in the evolutionary history of a race
- (3) Specific class of evolution
- (4) Not related to evolution

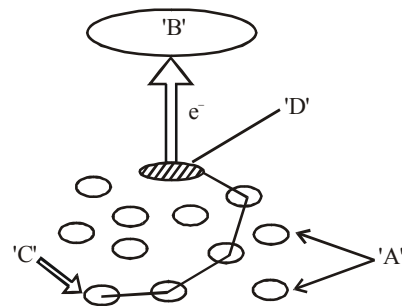
161. Which one is not the base of Whittaker's classification ?

- (1) Cell structure
- (2) Thallus organization
- (3) Mode of nutrition
- (4) Type of flagella

162. The date of reference for the correct scientific nomenclature of animals is :-

- (1) 1 May, 1753
- (2) 1 August, 1758
- (3) 1 May, 1758
- (4) 1 August, 1753

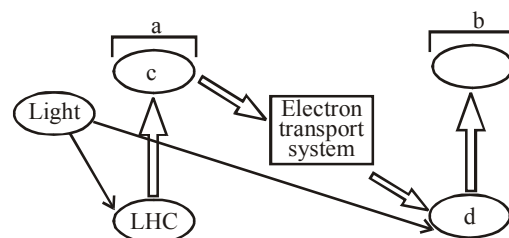
163.



The photosystem

Choose the correct match for A to D :-

- (1) A = photons, B = Reaction centre, C = Primary acceptor, D = Accessory pigment
- (2) A = Antenna molecules, B = Primary acceptor, C = photons, D = Reaction centre
- (3) A = Primary acceptor, B = Photons, C = Reaction centre, D = Antenna molecule
- (4) A = Antenna molecules, B = Reaction centre, C = Photons, D = Primary acceptor



164.

Choose the correct labelling for a to d -

- (1) a-PS-I, b-PS-II, c-electron acceptor of PS-II, d-LHC of PS-I
- (2) a-PS-II, b-PS-I, c-electron acceptor of PS-II, d-LHC of PS-II
- (3) a-PS-II, b-PS-I, c-electron acceptor of PS-II, d-LHC
- (4) a-PS-I, b-PS-II, c-electron acceptor of PS-II, d-LHC of PS-II

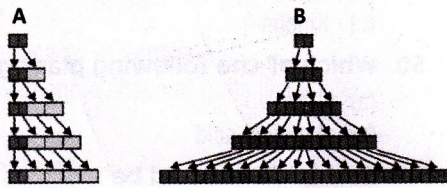
165. Mn is required in :

- (1) Chlorophyll-a
- (2) Nucleic acid synthesis
- (3) Plant cell wall
- (4) Photolysis of water during photosynthesis

166. Site of dark reaction in chloroplast is :

- (1) Thylakoids (2) Grana
- (3) Interganal lamellae (4) Stroma

167. Two types of growth are shown in the figures.
Select the correct statement :



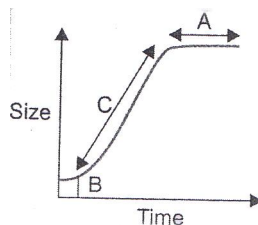
- (1) A is arithmetic growth curve is linear
- (2) B is arithmetic growth and growth curve is linear
- (3) A is geometric growth and growth curve is sigmoid
- (4) B is geometric growth and growth curve is linear

168. Match the following and choose the correct combination from the option given :

	Column-A (Growth Regulator)		Column-B (Action)
A	Absciscic acid	I	Delays leaf senescence
B	Ethylene	II	Inhibits seed germination
C	Cytokinin	III	Herbicide removal
D	Auxin	IV	Hastens fruit ripening

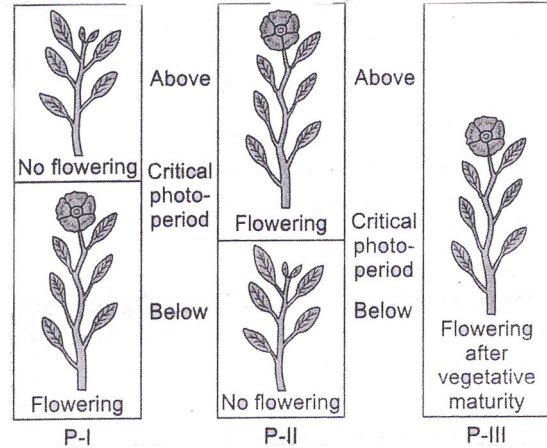
- (1) A – II, B – IV, C – I, D – III
- (2) A – I, B – II, C – III, D – IV
- (3) A – II, B – III, C – IV, D – I
- (4) A – II, B – I, C – III, D – IV

169. The picture below shows a graph drawn on the parameters of growth versus time, A, B, C respectively represent :



	A	B	C
1	Exponential phase	Log phase	Steady state phase
2	Steady state phase	Lag phase	Log phase
3	Log phase	Steady state phase	Logarithmic phase
4	Log phase	Lag phase	Steady state phase

170. The picture below shows three different types of plants (marked as P-I, P-II and P-III) which flower on the basis of their critical photoperiod. Now identify these plants (P-I, II and III) :



- (1) P-I = Long day plant; P-II = Short day plant; P-III = Day neutral plant
 - (2) P-I = Short day plant; P-II = Long day plant; P-III = Day neutral plant
 - (3) P-I = Short day plant; P-II = Short day plant; P-III = Day neutral plant
 - (4) P-I = Long day plant; P-II = Long day plant; P-III = Day neutral plant
171. FADH_2 is produced during the following reaction :-
- (1) Succinic acid to fumaric acid
 - (2) Fumaric acid to malic acid
 - (3) Succinyl Co-A to succinic acid
 - (4) Isocitric acid to oxaloacetic acid

172. A solution is different from pure water in having –

- (1) Low ψ_w
- (2) Less DPD
- (3) Less ψ_p
- (4) Less OP

173. Plasmolysis occurs when water moves out of cell and the cell membrane of a plant cell shrink away from it's cell wall. The water is drawn out of the cell from

- (1) Cytoplasm only
- (2) Wall
- (3) Vacuole
- (4) Both (1) & (3)

- 174.** Essential elements as components of energy related chemical compound in plants is/are
 (1) P
 (2) Mg
 (3) Mo
 (4) More than one option is correct
- 175.** Linnaeus in his book "Species plantarum" (1753) proposed scientific names and Latin diagnosis of *Solanum tuberosum*, *Mangifera indica*, *Saraca indica*, *Tamarindus indicus*, *Rosa indica* and *Solanum nigrum*. In the above case, how many species and genera respectively described by Linnaeus
 (1) 7, 5 (2) 6, 5
 (3) 5, 5 (4) 4, 5
- 176.** Neo systematics aims at :
 (1) Delimiting various taxa of organisms and establishing their relationship
 (2) Identification and arrangement of organisms on the basis of their cytological characteristics
 (3) The classification of organisms based on broad morphological characters
 (4) The classification of organisms based on their evolutionary history and establishing their phylogeny on the totality of various parameter from all field of studies
- 177.** Taxonomic aid which contains information on any one taxon, called :-
 (1) Herbarium (2) Zoo
 (3) Botanical garden (4) Monographs
- 178.** Match the following & choose the correct option:
 (A) Family (i) *Mangifera indica*
 (B) Kingdom (ii) Solonales
 (C) Order (iii) Solanum
 (D) Species (iv) Plantae
 (E) Genus (v) Malvaceae
 (1) (i)–D, (ii)–C, (iii)–E, (iv)–B, (v)–A
 (2) (i)–E, (ii)–D, (iii)–B, (iv)–A, (v)–C
 (3) (i)–D, (ii)–E, (iii)–B, (iv)–A, (v)–C
 (4) (i)–E, (ii)–C, (iii)–B, (iv)–A, (v)–D
- 179.** Find the **correct** match
 (A) Botanical garden – Have collection of living plant.
 (B) Zoological parks – Have collection of living animal
 (C) Biological museum – Have collection of preserved animals only
 (D) Taxonomical keys– Are used for identification of plants only
 (1) A & B (2) B & C
 (3) C & D (4) A & D
- 180.** Linnaeus system of plant classification is
 (1) Artificial
 (2) Natural
 (3) Phylogenetic
 (4) None of the above